Elimination



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Alterations in Urinary Elimination

- Dietary considerations:
  - Color of urine could be related to intake
- Age
  - Kidney function declines with age
  - Lose bladder tone
- Anatomy
- Privacy
- Medications
  - Diuretics
- Dehydration



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consumption of foods such as blackberries, beets, and rhubarb may turn urine a reddish color, so that it appears to be blood. Certain food dyes can also pass through the urine, turning it blue or green. Urine that is dark brown may occur after eating aloe or fava beans. Asparagus can change and increase the urine's odor.

•Antihistamines

•Alpha-blockers

•Anticholinergics/antispasmodics

•Antidepressants

•Antidiuretics

•Antihypertensives

•Antiparkinsonian medications

- Antipsychotics
- Decongestants
- Diuretics
- Estrogens
- Over-the-counter cold medications
- Narcotic pain relievers
- Psychotherapeutics

Urinary Assessment Terms



- Dysuria
- Frequency
- Polyuria
- Oliguria
- Nocturia
- Hematuria
- Anuria

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Urgency is immediate desire to void

Dysuria is painful urination

Frequency is voiding more than 8 times a day

Polyuria is excessive urination

Oliguria is diminished urination

Nocturia is urination at night

Hematuria is blood in the urine

Anuria means no urine output

Types of Urinary Incontinence

Stress incontinence: coughing, sneezing, laughing, or physical activity that increases pressure on the bladder
Urge incontinence: strong urge to urinate, but leaking occurs before getting to the toilet
Reflex incontinence: urinary leakage as a result of nerve damage
Overflow incontinence: incomplete bladder emptying which results in the bladder overfilling when full
Functional incontinence: physical inability to reach the toilet in time
Bedwetting (nocturnal enuresis)



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Urinary incontinence (UI) is the inability to control urination, resulting in the involuntary passage of urine. It can be caused by many factors. This common bladder control problem affects the retention, or holding, and releasing of urine from the bladder.

Interventions for Incontinence

- Lifestyle modifications
  - Limit causing agents (ex caffeine or alcohol)
- Pelvic floor exercises
- Medications
- Medical devices
- Surgery
- Monitor surrounding skin
- Utilize products to pull urine away from skin
- Skin barrier and cleansing products



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Urinary Retention

- Urinary retention occurs when the bladder is not able to fully empty.
- Symptoms of urinary retention include:
  - Inability to urinate
  - Pain
  - Abdominal distention
  - Urinary frequency
  - Urinary hesitancy
  - Weak or slow urine stream
  - Urinary leakage.
- Intervention based on source of problem



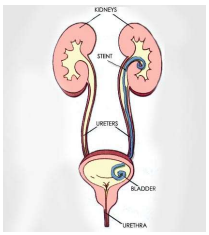
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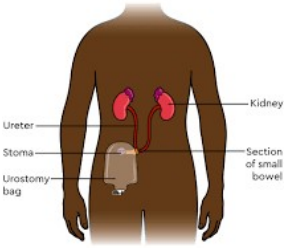
Urinary retention is a condition in which the bladder does not completely empty with urination. Health problems such as prostate enlargement or a cystocele (prolapsed bladder) can prevent urine from leaving the bladder fully.

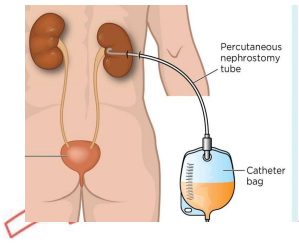
If not addressed, urinary retention can cause urinary tract infections, bladder damage, kidney damage, and urinary incontinence.

Urinary retention:
Maye be caused by BPH in mean, renal calculi, infection
Inability to partially or completely empty the bladder
Postvoid residual (PVR)

Urinary Diversions







Urinary Catheterization

Ureteral stent

- Can be temporary
- Allow passage of urine through blocked ureter

Urostomy

- Ileal conduit
- Surgically created diversion that uses small intestine
- Piece of small intestine removed and repositioned to attach to ureters and allow urine to pass

Nephrostomy

- Drain directly from kidney into pouch

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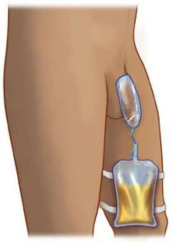
Blocked ureter can occur - many reasons, such as obstruction from a stone or mass, postoperative scar tissue, or inflammation and swelling from an infection.

Nephrostomy - A tube is surgically inserted through the skin on the back and into the kidney. This procedure is usually performed following the removal of kidney stones. This type of diversion is usually temporary and is removed once the kidney has healed.

Urinary Devices/Interventions

- Urinal and bedpan
  - Useful when client has limited mobility
  - Female vs male urinal
- Bladder scanning
  - Ultrasound device used to estimate amount of urine remaining in bladder
  - Reduce need for unnecessary catheterizations
- External catheters
  - Condom catheter
  - Purewick





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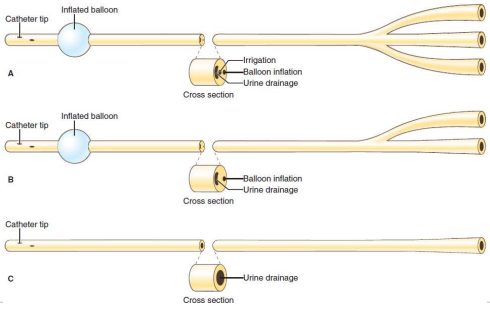
Purewick – continuous suction. Start at 40 and go up if needed. Not recommended to exceed 65mmHg

When and why would we bladder scan?

Catheterization

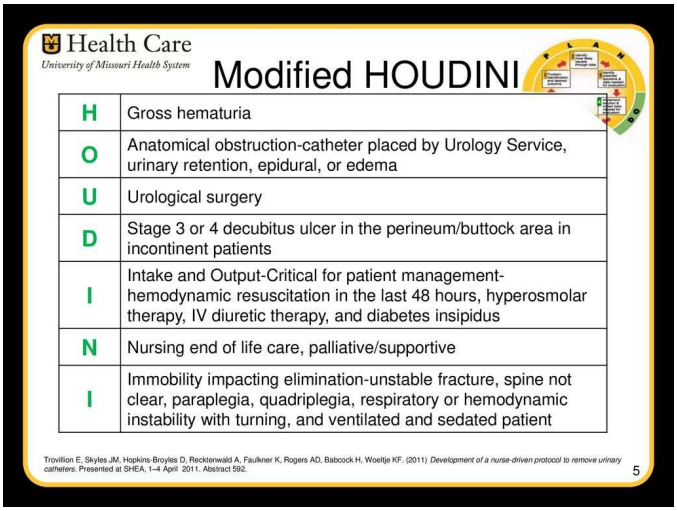
- Requires physician order
- Sterile procedure
- 14-16 French (Fr) tube
- Intermittent vs Indwelling
- Insertion of tube
- Drainage bag hangs below level of bladder
  - On non-movable part of bed
- Empty bag when about half full
- Check for kinks if no drainage
- Assess for removal





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Catheter Removal



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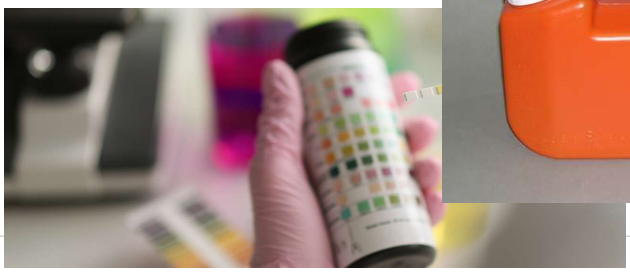
I – intake and output when needed to be strict. Especially for critically ill hemodynamically unstable clients

I – immobility – example, new spinal cord injury, unstable fracture

Urinary Diagnostics

- Postvoid Residual (PVR)
  - Measures amount of urine in bladder after voiding
- Cystoscopy
  - Optical instrument used to view lining of urethra and bladder
- Pyelogram
  - Intravenous vs Retrograde
- Urinalysis
  - Clean catch
  - Sterile specimen
  - Random specimen
- Urine culture

- 24-hour urine collection
  - Uses special container for collection
  - Must remain refrigerated or on ice
  - Void first urine and start time
  - Collect all other voids within 24 hours





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Intravenous Pyelogram (IVP)
Contrast dye is injected intravenously, filtered by the kidneys, and excreted into the urinary tract. X-rays are taken in sequence as the contrast moves through the kidneys, ureters, and bladder, creating an image of the entire upper and lower urinary tract. Used to identify kidney stones, obstruction, tumors, or structural abnormalities.

Retrograde Pyelogram:
Contrast is injected directly into the ureter via a cystoscope (going up, or retrograde, rather than through the bloodstream). Used when IV contrast is contraindicated or when better visualization of the ureters or renal pelvis is needed. Often performed by urology.

Urinalysis is a lab test commonly used by health care providers to evaluate for a variety of disorders, including bladder infections or UTIs, kidney infections, kidney disease, and diabetes. This test includes visual examination of the urine, dipstick testing, and microscopic examination.

Clean catch - patient voids and discards a small amount of urine; continues voiding in a sterile specimen container to collect the urine; stops voiding into container;

removes container and continues voiding; then discards the last amount of urine in the bladder. The first small amount of urine voided helps to flush away any organisms near the meatus because the findings may be inaccurate if these organisms enter the specimen. In addition, it is generally thought that urine voided at midstream is most characteristic of the urine the body is producing.

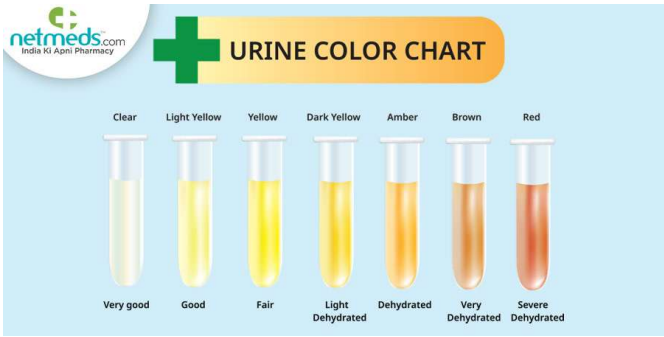
Sterile urine specimens may be obtained by catheterizing the patient's bladder or by taking the specimen from an indwelling catheter already in place

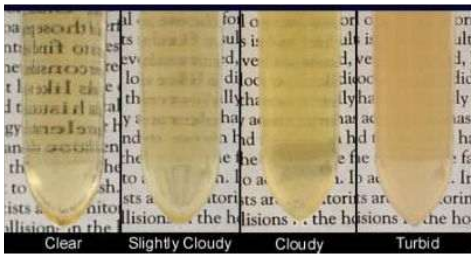
Random means just that – a random sample at anytime

Urine culture - A urine culture is used to evaluate urine for the presence of bacteria and yeast that may cause a UTI. This test is commonly ordered in addition to a urinalysis to confirm the presence of bacteria in urine revealed on the urine reagent strip.

24 hour - Another diagnostic test for the urinary system is 24-hour urine collection, in which urine is collected over a 24-hour period, placed in a special container, and refrigerated. The client can collect the urine at home after the nurse reviews how to perform the procedure. This test allows the provider to evaluate the amounts of chemicals that are in the urine over a 24-hour period and can be helpful in determining kidney function.

Urine Assessment





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Alterations in Elimination

- Alterations in bowel elimination are usually treatable or manageable.
- Nurses can assist the client with making lifestyle changes, provide bowel training, and provide the client with agents to stimulate a bowel movement.

- Dietary considerations
  - Diet can influence stool production
  - Certain foods make clients more prone to diarrhea
- Age
  - Stomach elasticity diminishes with age
    - Amount of food and rate of emptying change
  - Decrease in peristalsis and muscle tone
    - Increase in constipation
  - Inactivity, decreased fiber intake, and medications can impact bowels
- Medications
- Privacy
- Surgery



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These factors include the following foods and beverage, which are known to increase the risk of diarrhea:

•Alcohol

•Caffeinated beverages and foods

•Dairy (milk, cheese, cream, and ice cream)

•Foods that are high in fat or are highly greasy

•Beverages that contain fructose

•Spicy foods

•Apples, peaches, and pears

•Products that contain sweeteners, such as sorbitol, mannitol, xylitol, and maltitol

- Antacids containing aluminum and calcium
- Anticholinergics/antispasmodics
- Antiseizure
- Antidepressants
- Antihypertensives
- Antipsychotics
- Bile acid sequestrants
- Calcium channel blockers
- Diuretics
- Iron supplements
- Opiates

Constipation

- Fewer than three bowel movements in a week.
- A condition where the production of stool is slowed.
- Can result in dry and hard to pass bowel movements and give a sensation of incomplete emptying or passing of stool.
- Interventions:
  - Increased fluid intake
  - Stool softeners, suppositories, enemas
  - High-fiber diet
  - Exercise



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Constipation occurs when a client has infrequent bowel movements, often defined as fewer than three bowel movements in a week. Additionally, these stools are hard, lumpy, and difficult to pass.

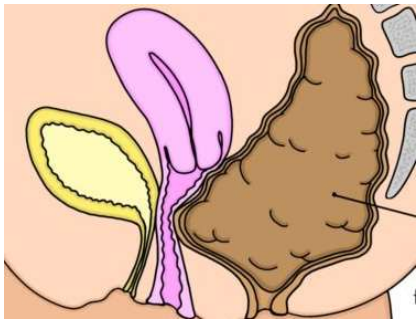
Manifestations that require medical attention include fever, bleeding from the GI tract, abdominal pain, vomiting, low back pain, and weight loss. If constipation is chronic, fecal impaction or obstruction can occur.

Interventions to treat constipation include a high-fiber diet, staying well hydrated, exercise, bowel training, and medications to soften stools.

Impaction

Impaction

- Stool gets stuck in rectum or colon
- Results from unrelieved constipation
- Manifestations:
  - Inability to have bowel movement
    - May have small amount of stool that can navigate around impaction
  - Abdominal or back pain
  - Nausea and/or vomiting
  - Difficulty urinating
- Intervention
  - Enema
  - NG tube
  - Manual disimpaction



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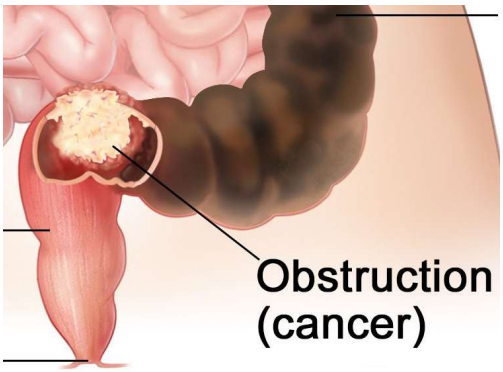
fecal impaction is a type of constipation where a large, hard mass of stool gets trapped in the colon or rectum. A bowel obstruction is a blockage in the small or large intestine that prevents the contents from passing through

The decision for hospitalization and surgical consultation is a powerful tool in identifying the obstruction's type, severity, and cause. The placement of a nasogastric (NG) tube insertion is a key intervention that relieves distention, nausea & vomiting through bowel decompression, allows for accurate emesis output assessment, and significantly decreases the risk for aspiration.

Bowel Obstruction

- Partial or complete blockage of small or large bowel
- May result from unrelieved impaction or other blockage such as a tumor
- Can be life threatening
  - Need prompt diagnosis
- Manifestations:
  - Nausea and vomiting
  - Abdominal pain and distention
  - Severe constipation
    - May only produce small amount of stool that can pass obstruction
- Interventions:
  - NG tube
  - surgery

Obstruction



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Diarrhea

- Frequent loose, water stools
- Can be acute, persistent, or chronic
- Often urgent need to defecate
- May have cramping, bloating, and nausea
- Risk factors:
  - Infection
  - Medication use
  - GI disorders
  - Diet
- Interventions
  - Imodium
  - Monitor hydration status
  - Avoid known irritants

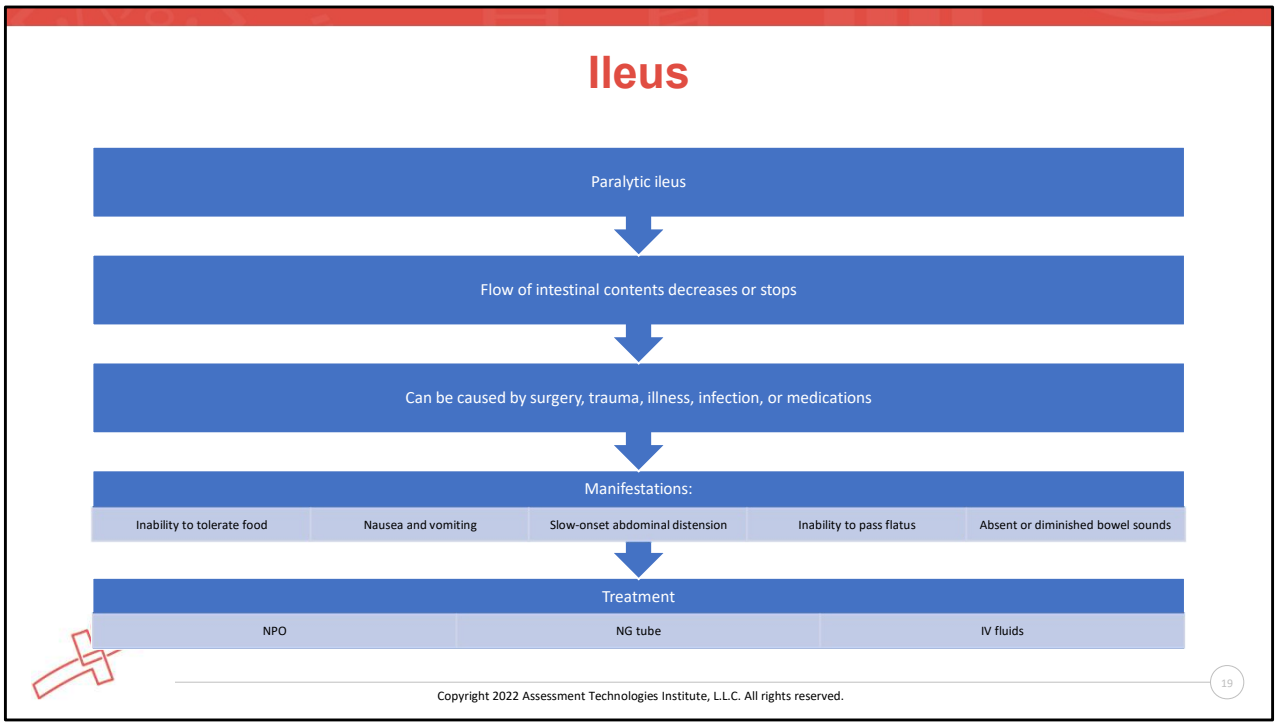


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It can be acute, lasting about 1 to 2 days; persistent, lasting longer than 2 weeks but less than 4 weeks; or chronic, lasting longer than 4 weeks.

If diarrhea is persistent, dehydration and malabsorption can occur, and both can be life-threatening if severe. In adults, symptoms that require medical follow-up would include fever of 102°F or higher, diarrhea lasting longer than 2 days or six or more bowel movements a day, abdominal pain, or the presence of blood or black feces.

Interventions are aimed at removing irritating factors and ensuring rehydration. If the client is experiencing an acute case of diarrhea, over-the-counter medications such as loperamide or bismuth subsalicylate may be used. If GI infection is the cause, antibiotics and/or probiotics will be prescribed to help restore the intestinal flora and re-create the usual GI environment.



Identifying and addressing the underlying cause can potentially decrease the longevity of the ileus. However, if the client cannot tolerate an oral diet after seven days, the healthcare provider should prescribe total parenteral.

Clostridium Difficile (C. diff)

- Bacteria that causes an infection of the colon
- Risk factors
  - Antibiotic use
  - Weakened immune system
  - Older than 65 years
  - Recent hospitalization
  - Previous C. diff infection
- Manifestations
  - Diarrhea
    - Often watery, yellow, mucous
    - Foul smelling
  - Abdominal pain or cramps
  - Loss of appetite
  - Nausea
  - Rapid heart rate



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•**Antibiotics**: Being exposed to antibiotics, especially fluoroquinolones, clindamycin, carbapenems, or third or fourth generation cephalosporins. The risk of getting C. diff is up to 10 times higher while taking antibiotics or within three months after finishing them.

•**Weakened immune system**: Having a weakened immune system due to a medical condition or treatment, such as HIV/AIDS, cancer, or taking immunosuppressive drugs for an organ transplant

Clostridium Difficile (C. diff) cont.

- Complications
  - Dehydration
    - Can lead to kidney failure
  - Skin irritation
  - Discomfort
  - Sepsis
- Diagnostics
  - Stool culture
- Interventions
  - Contact precautions
  - Antibiotics
    - Vancomycin or Flagyl
  - Pain medications
  - IV fluids
  - Clear diet



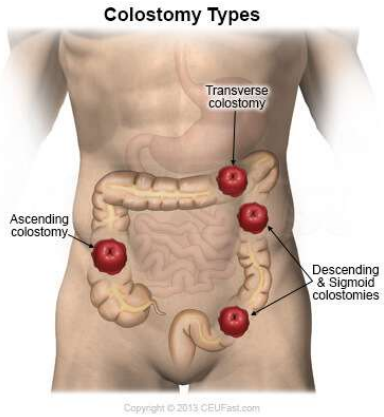
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•**Antibiotics**: Being exposed to antibiotics, especially fluoroquinolones, clindamycin, carbapenems, or third or fourth generation cephalosporins. The risk of getting C. diff is up to 10 times higher while taking antibiotics or within three months after finishing them.

•**Weakened immune system**: Having a weakened immune system due to a medical condition or treatment, such as HIV/AIDS, cancer, or taking immunosuppressive drugs for an organ transplant

Ostomies

- Can be temporary or permanent diversion of intestinal contents
- Ileostomy
  - Uses small intestine
  - Redirect ileum through created opening (stoma)
  - Stool will be liquid
- Colostomy
  - Part of colon is redirected
  - Can be anywhere in colon
  - Stool depends on where ostomy is placed in colon



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Ascending is still liquid, but thicker.

Transverse is soft stool, not liquid anymore.

Descending is thick, almost formed.

Sigmoid is formed stool.

Assessment of Site & Stoma

- Assess ostomy site for skin breakdown or irritation
- Stoma should be moist and pink or red in color
- Blue stoma indicates poor perfusion
  - Call the physician
  - May be able to save without surgery
- Black stoma indicates tissue death
  - Call the physician
  - Likely needs to return to the OR









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Bowel Interventions

Enemas

- Instillation of liquid solution into GI tract
- Type of solution dependent on physician order
  - Tap water
  - Normal Saline
  - Hypertonic solution (Fleets)
  - Soapsuds
  - Oil-retention
- Clean technique
  - Lay client on left side
  - Insert tube 3-4 inches
  - If cramping lower height of bag



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Tap water is hypotonic and exerts an osmotic pressure lower than fluid in interstitial spaces. After infusion into the colon, tap water escapes from the bowel lumen into interstitial spaces. The net movement of water is low. The infused volume stimulates defecation before large amounts of water leave the bowel. Use caution if ordered to repeat tap-water enemas because water toxicity or circulatory overload could develop if the body absorbs large amounts of water.

Physiologically, normal saline is the safest solution to use because it exerts the same osmotic pressure as fluids in interstitial spaces surrounding the bowel. The volume of infused saline stimulates peristalsis. Giving saline enemas lessens the danger of excess fluid absorption.

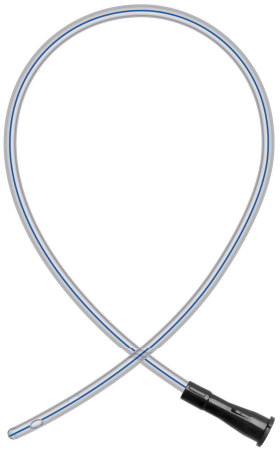
Hypertonic solutions (fleets) infused into the bowel exert osmotic pressure that pulls fluids out of interstitial spaces. The colon fills with fluid and the resultant distention promotes defecation. Patients unable to tolerate large volumes of fluid benefit most from this type of enema, which is by design low volume. This type of enema is contraindicated for patients who are dehydrated and young infants. The commercially prepared Fleet enema is the most common.

You add soapsuds to tap water or saline to create the effect of intestinal irritation to stimulate peristalsis. Use only pure castile soap that comes in a liquid form that is included in most soapsuds enema kits. Use soapsuds enemas with caution in pregnant women and older adults because they could cause electrolyte imbalance or damage to the intestinal mucosa.

Oil-retention enemas lubricate the feces in the rectum and colon. The feces absorb the oil and become softer and easier to pass. To enhance action of the oil, the patient retains the enema for several hours if possible.

Rectal tube

- Short term intervention
  - Facilitate passage of flatus or liquid stool
- Used when distension causes discomfort or difficulty breathing
- Position client left lateral
- Insert tube 3-4 inches
- Stays in no longer than 20-30 minutes
- Document output and client response
- Goal is relief of distension



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Fecal Management System

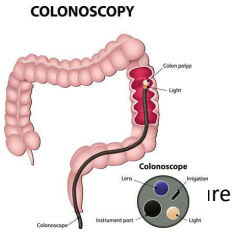


- Indwelling device diverts semi-liquid to liquid stool into collection system
  - Prevents stool from sitting on skin
- Contraindications:
  - Previous rectal or anal surgery
  - Rectal or anal stricture (narrowing)
  - Severe hemorrhoids
  - Rectal tumor
  - Bowel obstruction or impaction
- Balloon inflated with water
  - Gentle tug after insertion to ensure placement
- Check balloon integrity each shift
- Monitor output and surrounding skin

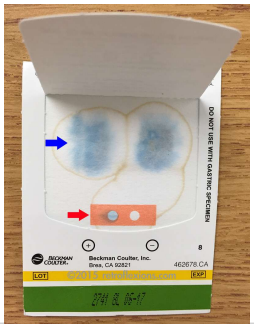
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Bowel Diagnostics

- Colonoscopy
  - Screening procedure
- Upper and Lower GI series
  - Barium given either orally or rectally
  - X-rays taken to examine tract
- Fecal Occult Blood test
  - Identify if blood is in stool
  - Can delegate collection of sample
  - Cannot delegate placing drops or reading test
- Stool culture
  - Identify any organisms in stool

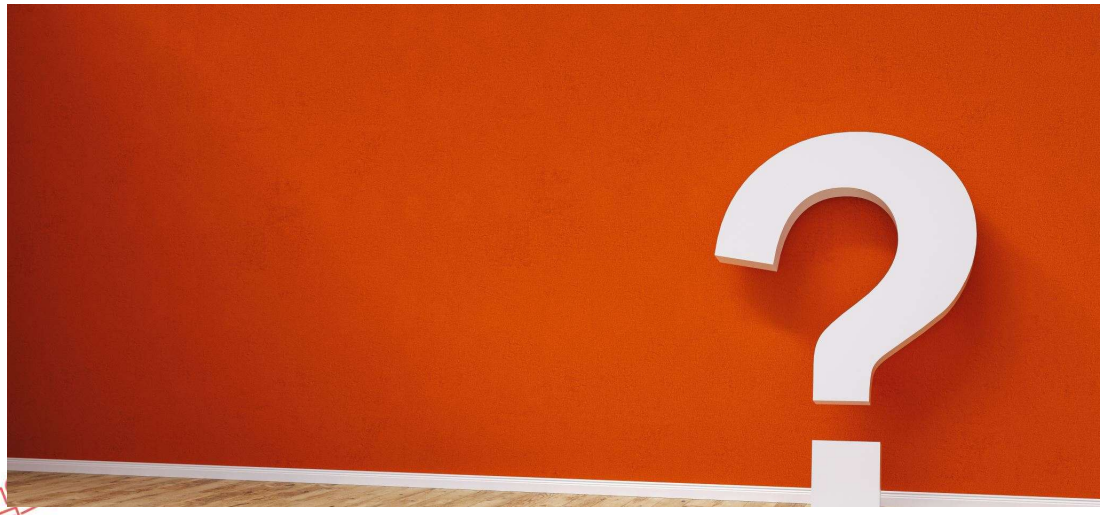






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